

CHEAT SHEET

Building AI Agents with Claude

A condensed reference of every module — the big idea and key
takeaway, in one sitting.

32 modules · From Hello World to Autonomous Production Systems

MODULE

TAKEAWAY · The LLM is the brain, tools are the hands, and the loop keeps it working until the job is done.

MODULE 1 OF 30

Track 1: Foundations

The LLM Mental Model

Understand what a Large Language Model really does, how it processes your text, and why this mental model changes everything you build.

A Large Language Model like Claude is the world's most well-read autocomplete. It doesn't "understand" language the way humans do — it predicts the most likely next token based on patterns learned from billions of documents.

TAKEAWAY · Think of Claude as a "thinker, not a calculator."

MODULE 02 OF 30

Track 1: Foundations

Tokens — The Atoms of AI Communication

Every cost, every limit, every performance decision traces back to tokens. Learn what they are and why they matter.

Every interaction with Claude — every cost, every limit, every performance decision — traces back to tokens. Tokens are subword chunks that break text into reusable pieces, like syllables help children learn to read.

TAKEAWAY · A customer-support agent processing 50,000 calls/day at 2,500 tokens each costs ~\$700/day on Sonnet.

MODULE

TAKEAWAY · The system prompt is your agent's operating manual — define role, rules, and output format clearly.

MODULE

TAKEAWAY · Context engineering is the discipline of choosing what the model sees on each turn — under a finite token budget.

MODULE 4 OF 30

Track 1: Foundations

Structured Output & Parsing

Agents produce data your code must parse, validate, and act on. Learn how to get reliable, structured output from Claude every time.

TAKEAWAY · Structured output is the contract between AI and your code. Use tool use with JSON Schema to force Claude into predictable data shapes. Always validate with Pydantic or Zod. Always implement error-aware retry (send the error back to Claude).

Function Calling — Giving Claude Hands

The core mechanism that turns a chatbot into an agent: Claude decides what to do, your code does it.

TAKEAWAY · Function calling is the bridge between AI reasoning and real-world action. Claude decides what to do; your code does it.

MODULE

TAKEAWAY · Every multi-tool agent balances three forces:

MODULE

TAKEAWAY · MCP turns the $N \times M$ integration problem into $N + M$. Instead of building a custom adapter for every model-tool combination, you build one MCP server per tool and one MCP client per AI app. They discover each other's capabilities automatically through a standard handshake.

MODULE

TAKEAWAY · Claude has zero memory between API calls. Every "memory" the model has comes from the messages array your code sends. A conversation manager is the essential middleware that decides what Claude remembers and what gets compressed or dropped.

MODULE

TAKEAWAY · You search your documents for relevant pieces, paste them into the prompt, and let Claude answer from that context. No model changes needed, no retraining, no fine-tuning.

MODULE

TAKEAWAY · Hybrid search fixes keyword blindness -- when vector search misses exact terms like invoice numbers or product codes.

MODULE

TAKEAWAY · The key insight of multi-layer memory is that different kinds of information need different retrieval strategies and retention policies. Working memory is fast and ephemeral (cleared per task). Episodic memory is persistent and searched by semantic similarity. Procedural memory is persistent and matched by task type.

MODULE

TAKEAWAY · Claude's API does this natively — text blocks are reasoning, tool_use blocks are actions, tool_result messages are observations. You just write the loop and the tool handlers.

MODULE

TAKEAWAY · Planning transforms agents from reactive tool-callers into strategic problem-solvers. The key pattern: classify intent first (skip planning for simple tasks), decompose complex goals into a DAG of sub-tasks, execute independent tasks in parallel, and synthesize results.

MODULE

TAKEAWAY · Multi-agent systems split complex tasks across focused specialists coordinated by a supervisor. The recommended architecture is hub-and-spoke : one coordinator delegating to workers, each with a small tool set and focused prompt.

MODULE

TAKEAWAY · The pattern is simple: Claude writes code, a sandboxed interpreter runs it, results come back. Always sandbox with Docker: no network , memory limits , read-only filesystem , timeout .

MODULE

TAKEAWAY · That's the key architectural insight of this entire module. Each subagent gets its own focused prompt, narrow tool set, and blank context window.

MODULE

TAKEAWAY · Order them cheapest-first : rate limiting (free), PII regex (sub-millisecond), injection classification (~200ms API call), schema validation (free). This ordering minimizes cost and latency — most blocked requests never trigger an API call.

MODULE

TAKEAWAY · Format validation is free — just local JSON parsing. Hallucination detection costs ~\$0.003 per check — trivially cheap compared to acting on fabricated data.

MODULE

TAKEAWAY · Use Claude-as-judge with detailed rubrics to score each output on task completion, tool accuracy, and response quality. Require reasoning before scores for consistency.

MODULE

TAKEAWAY · Capture traces (complete execution records), spans (timed units of work), and structured logs (machine-parseable JSON) for every request.

MODULE

TAKEAWAY · Without dashboards you can't detect problems. Without tiered alerting you drown in noise or miss emergencies. Without feedback loops at multiple timescales, you fix crises but never prevent them.

MODULE

TAKEAWAY · SSE streams tokens with zero polling waste. Docker slim images give portable, secure containers. Cloud Run provides SSE support, 60-min timeouts, and scale-to-zero billing.

MODULE

TAKEAWAY · Caching eliminates redundant API calls. Model routing sends 60-70% of traffic to Haiku at 12x lower cost. Token compression tackles the hidden cost multipliers.

MODULE

TAKEAWAY · The lab built topology 1: sync API — Docker, Cloud Run, Lambda. Same code, same agent, three deploys. Use this when requests finish in under 30 seconds.

MODULE

TAKEAWAY · The five-phase approach (Requirements, Architecture, Build, Evaluate, Harden) is the difference between agents that work on test day and agents that work in production.

MODULE

TAKEAWAY · The agent frontier is defined by four converging trends:

MODULE

TAKEAWAY · Claude Code mastery comes from three practices:

MODULE

TAKEAWAY · Three rules for the certification exam and production systems:

MODULE

TAKEAWAY · The single principle that unlocks most exam questions:

MODULE

TAKEAWAY · Domain 5.5/5.6 is the biggest gap in most prep — and the fix isn't a better prompt. It's a stricter output schema.